

★★ function

In C++ function is used to perform any specific task.

It is used for code reusability.

It has three parts

- ① function declaration
- ② function Body
- ③ function call

function Declaration

written type function name parameter passing
 ex- $\uparrow$ $\uparrow$ $\uparrow$
 int sum (inta, intb)

- function declare above main function.
- function body written after main function
- function calling in main function

* Programme of sum of two number

```
#include <iostream>
```

```
using namespace std;
```

funcn
declairn ← `int sum (inta, intb);`

```
int main()
```

```
{
```

```
  int a, b, c;
```

```
  cin >> a >> b;
```

funcn
calling ← `c = sum (a, b);`

```
}
```

funcn
body ← `int sum (int x, int y)`

```
{
```

```
  int z;
```

```
  z = x + y;
```

```
  return z;
```

```
}
```

Programme

#include <iostream>

using namespace std;

int sum(int a, int b);

int main()

{

int a, b, c, d, e;

Cin >> a >> b;

c = sum(5, 10);

d = sum(6, 12);

e = sum(10, 19);

Cout << sum c << sum d << sum e;

}

int sum(int x, int y)

{

int z;

z = x + y;

return z;

}

funcⁿ
declⁿ ←
①

funcⁿ
callⁿ ←
③

funcⁿ
body ←
②

Q. write a programme to swap two integer no. by using function

#include <iostream>

using namespace std;

void swap(int a, int b);

int main()

{

int a, b;

Cin >> a >> b;

Cout << "before swapping" << a << b;

swap(a, b);

return 0;

}

funcⁿ
declⁿ ←
①

funcⁿ
call ←
③

funcn ← void swap (int x, int y)
body
② {

int t;

t = x;

x = y;

y = t;

cout << "after swaping" << x << y;

}

** Structure

- It is kind of user-define datatype
- It is collection of different type of datatype under a single name
- It has two part
 - 1) data member
 - 2) member function
- we can create structure by using ~~structure~~ keyword
- Structure written above main function.

Programme

```
#include <iostream>
```

```
using namespace std;
```

```
struct Student
```

```
{
```

```
int age;
```

```
float marks;
```

```
string name;
```

```
};
```

```
int main()
```

```
{
```

```
int a; float b, string c;
```

```
Student s1, s2, s3;
```

S.age = a;

S.Float = b;

S.Name = c;

cout << S.age;

cout << S.marks;

cout << S.Name;

}

Array

→ Array is a list of variables of similar type

ex-

{1, 2, 3, 4} ✓

{'c', 'f', 'h', 'i'} ✓

{'d', 3, 'f', 4.3} ✗

→ Array have Contiguous memory

→ ~~initialization~~ Array

→ consecutive block of memory

→ Declaration of Array

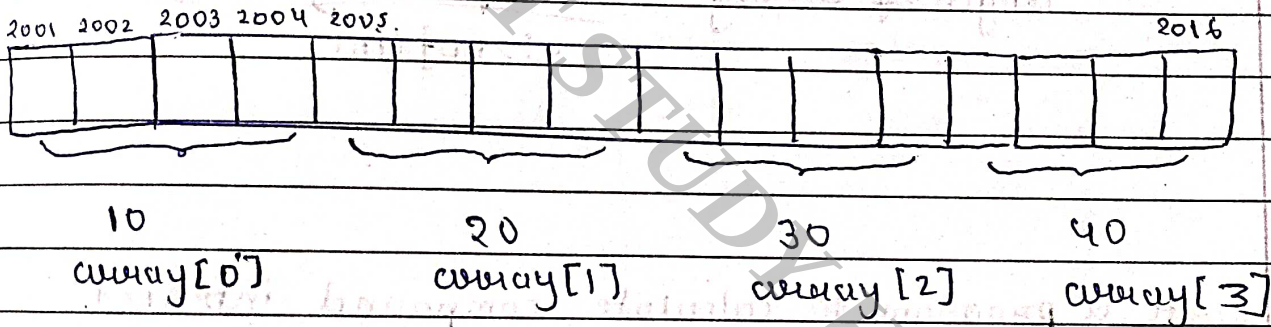
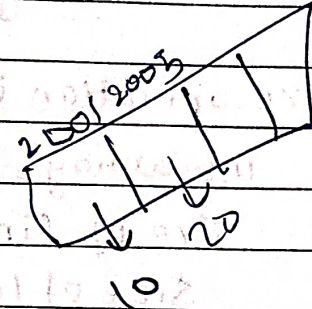
datatype array Name [size]

→ initialization of Array

ex- `int array[4] = {10, 20, 30, 40};`

int - 4 bytes

array = $4 \times 4 \text{ bytes} = 16 \text{ bytes}$



To access variable in array we use index
index start from 0, index determine location of every element

ex- `#include <iostream>`

`using namespace std;`

`int main()`

`{`

`int n;`

`cin >> n;`

`int array[n];`

`for (int i=0; i<n; i++)`

`{`

`cin >> array[i];`

`}`

`for (int i=0; i<n; i++)`

`{`

`cout << array[i] << " ";`

`}`

`return 0;`

`}`

★ 1D Array

⇒ Accessing element in 1D array

- based on 0-index

index (last element) = size - 1

ex. `int array[] = {1, 2, 3, 4}`

`array[0] = 1`

`array[1] = 2`

`array[2] = 3`

`array[3] = 4`

Size operation in array

`int array[] = {1, 2, 3, 4}`

Size of (int) = 4

Size of (array) = $4 \times 4 = 16$ byte

length of array = $\frac{\text{size of (array)}}{\text{size of (int)}} = \frac{16}{4} = 4$

Date: 5/3/24

Date
Page

Array

Array is derived datatype

Array is used to hold the data that are of similar type.

Array is the collection of similar type of data.

Q- Write a

There are two types of array :-

1. One Dimensional array

(1-D array)

2. Multidimensional array
(2-D, 3-D, 4-D Array)

Q- Write a program to insert elements in array and print them. (array).

```
int main ()
```

```
{
```

```
    int a[5];
```

```
    for (int i = 0; i < 5; i++)
```

```
    {
```

```
        cin >> a[i];
```

```
    }
```

```
    for (
```

```
    {
```

```
        cout << a[i];
```

```
    }
```

```
}
```

7	6	8	1	0
---	---	---	---	---

Always start from 0

Q- Write a program to calculate sum of array elements and find out average of all elements

```
int main ()
{
    int a[5];
    for (int i = 0; i < 5, i++)
    {
        cin >> a[i];
        // for (int i = 0; i < 5, i++)
        // cout << a[0] + a[1] + a[2] + a[3] + a[4];
        sum = sum + a[i];
    }
}
```

float a[5];
char a[6];
const
'A' 'b'

```
int main ()
{
    int a[5], sum = 0, avg;
    for (int i = 0; i < 5, i++)
    {
        cin >> a[i];
        sum = sum + a[i];
        cout << sum;
    }
    avg = sum / 5;
    cout << avg;
}
```

return